

ASSESSMENT TOOL 2: Logic Reasoning & Mapping Exercise

COURSE NAME: ARTIFICIAL INTELLIGENCE COURSE CODE: MLA01
AND EXPERT SYSTEMS

COURSE OUTCOME COVERED

CO3: *Implement propositional and first-order logic using resolution, unification, and inference techniques for knowledge representation and reasoning.* (BTL 3)

Assessment Tool Weightage: 15%

Total Marks: 100

Time: 60 Mins

No. of Questions: 5

Annexure A

Logic Reasoning & Mapping Exercise

Code of Conduct:

I Sk. Fawaz Ahamad / 192425390 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Sk. Fawaz Ahamad

Criterion	Weightage	Marks Obtained
1. Problem Analysis and Knowledge Identification	15%	
2. Logical Representation and Mapping	20%	
3. Predicates, Variables, Constants and Quantifiers	15%	
4. Logical Connectives and Formula Construction	10%	
5. Inference and Logical Reasoning	15%	
6. Unification and Substitution	10%	
7. Resolution and Proof Construction	10%	
8. Mapping Diagram / Clarity and Presentation	5%	
Total	100%	

ANSWER ANY ONE OF THE FOLLOWING

Q.No	Use Case Scenario	
1	Smart Healthcare Diagnosis	A hospital uses an AI system to identify possible diseases from patient symptoms. Ravi has fever, cough, and body pain. Patients with fever and cough may have flu. Patients with flu and body pain may have viral infection. Patients with viral infection require medical attention. Tasks: (i) Identify facts, entities, predicates, variables, and constants. (ii) Represent the knowledge using Propositional Logic. (iii) Represent the knowledge using FOL with suitable quantifiers. (iv) Perform unification for two relevant predicates. (v) Apply resolution to prove whether Ravi has a viral infection. (vi) Show the complete inference sequence and final conclusion.
2	Student Academic Advising	A university AI system recommends academic support based on attendance and examination performance. Priya has attendance below 75% and has failed two courses. Students with low attendance require attendance counseling. Students who fail two or more courses require academic counseling. Students requiring both types of counseling need academic support. Tasks: (i) Identify the knowledge components. (ii) Convert the statements into Propositional Logic. (iii) Develop FOL representations using predicates and quantifiers. (iv) Identify suitable substitutions and perform unification. (v) Use resolution to prove whether Priya requires academic support. (vi) Explain the reasoning steps leading to the conclusion.

3	Intelligent Approval	Loan	A bank wants to use an AI knowledge-based system for preliminary loan approval. Arun has stable employment, sufficient income, a high credit score, and good repayment history. Customers with stable employment and sufficient income qualify for preliminary approval. Customers with a high credit score and preliminary approval are classified as low-risk. Low-risk customers with good repayment history can be recommended for loan approval. Tasks: (i) Identify entities, facts, predicates, and rules. (ii) Map the scenario into Propositional Logic. (iii) Represent the rules using FOL.
			(iv) Demonstrate unification and substitution. (v) Use resolution to prove or disprove LoanApproved(Arun). (vi) Provide the final inference with justification.
4	Vehicle Diagnosis	Fault	An automobile service center uses an expert system to diagnose vehicle faults. Car101 does not start and its headlights are dim. Vehicles that do not start and have dim headlights may have a battery problem. Vehicles with a battery problem require battery inspection. Vehicles requiring battery inspection are marked for service. Tasks: (i) Extract the facts and rules from the scenario. (ii) Represent them using Propositional Logic. (iii) Develop suitable FOL predicates and rules. (iv) Perform unification for the vehicle-related predicates. (v) Apply resolution to determine whether Car101 has a battery problem. (vi) Derive whether Car101 should be marked for service and explain the inference.

5	Cybersecurity Threat Detection	An organization uses an AI system to identify cyber threats. User101 has five failed login attempts and logged in from an unusual location. Multiple failed login attempts indicate a possible brute-force attack. A brute-force attack combined with an unusual login location indicates suspicious activity. Suspicious activity combined with unauthorized access to confidential files indicates a possible security breach. Tasks: (i) Identify the facts, entities, predicates, and rules. (ii) Map the knowledge into Propositional Logic. (iii) Represent the rules using FOL. (iv) Demonstrate unification and substitution. (v) Apply resolution to determine whether User101 is involved in a possible security breach. (vi) Present the complete reasoning chain and final inference.
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SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES (SIMATS)

School of Engineering

Department of Artificial Intelligence & Machine Learning

CO3-AT2

Knowledge Representation & Reasoning

CRITERION-WISE ANALYSIS REPORT

MLA0103 – Artificial Intelligence and Expert Systems

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Year	III Year
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1. ASSESSMENT CRITERIA

Criterion	Weightage	Marks Obtained
1. Problem Analysis and Knowledge Identification	15%	
2. Logical Representation and Mapping	20%	
3. Predicates, Variables, Constants and Quantifiers	15%	
4. Logical Connectives and Formula Construction	10%	
5. Inference and Logical Reasoning	15%	
6. Unification and Substitution	10%	
7. Resolution and Proof Construction	10%	
8. Mapping Diagram / Clarity and Presentation	5%	
Total	100%	

The following sections are organized using the exact criterion titles and weightages shown in the supplied assessment table. Each criterion is addressed with medium-length content, formal representations, examples, and supporting visuals where they improve clarity.

2. USE CASE SCENARIO

A hospital wants to develop an AI-based system that identifies possible illnesses from patient symptoms. In the given case, Ravi has fever, cough, and body pain. The knowledge base states that patients with fever and cough may have flu, patients with flu and body pain may have a viral infection, and patients with viral infection require medical attention.

Given Facts and Rules

Type	Statement	Logical Form
Fact	Ravi has fever.	Fever(ravi)
Fact	Ravi has cough.	Cough(ravi)
Fact	Ravi has body pain.	BodyPain(ravi)
Rule	Fever and cough imply flu.	$\text{Fever}(x) \wedge \text{Cough}(x) \rightarrow \text{Flu}(x)$
Rule	Flu and body pain imply viral infection.	$\text{Flu}(x) \wedge \text{BodyPain}(x) \rightarrow \text{ViralInfection}(x)$
Rule	Viral infection requires medical attention.	$\text{ViralInfection}(x) \rightarrow \text{MedicalAttention}(x)$

Main Tasks

1. Identify facts, entities, predicates, variables and constants.
2. Represent the knowledge using Propositional Logic.
3. Represent the knowledge using First-Order Logic with suitable quantifiers.
4. Perform unification for two relevant predicates.
5. Apply resolution to prove whether Ravi has a viral infection.
6. Show the complete inference sequence and final conclusion.

Figure 1 — Smart Healthcare Knowledge Flow

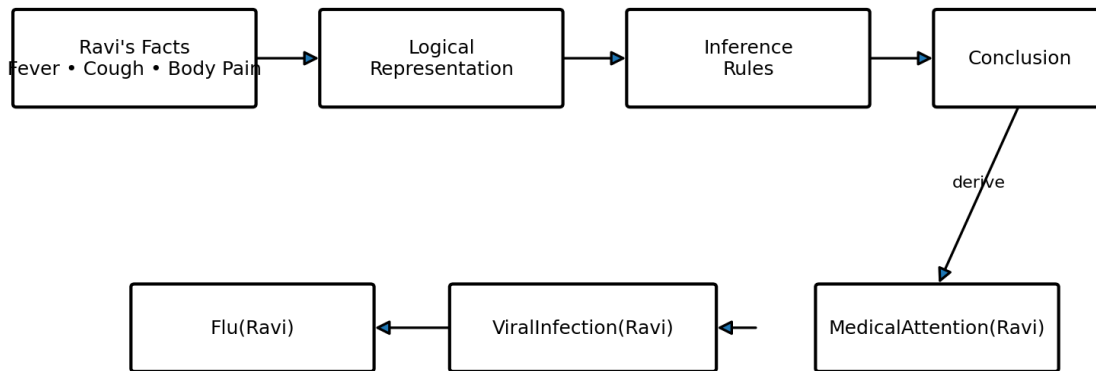


Figure 1. Problem domain: symptoms are transformed into logical facts and conclusions.

3. 1. Problem Analysis and Knowledge Identification

The problem represents a knowledge-based healthcare diagnosis scenario in which the system reasons from known patient symptoms to possible conditions. The first requirement is to identify the information explicitly provided and separate it from conclusions that must be inferred. Ravi is the specific patient, while fever, cough and body pain are the initial observable facts. The rules supplied in the problem define how these symptoms can be combined to derive flu and then viral infection.

The knowledge identification stage therefore establishes the basic components of the knowledge base: the patient constant ravi, symptom predicates, disease predicates and the medical-attention predicate. The target query is whether `ViralInfection(ravi)` can be logically derived. This clear problem analysis ensures that the reasoning process remains limited to the facts and rules provided in the question.

Identified Knowledge

- Entity: Ravi, the patient described in the scenario.
- Constant: ravi represents the specific individual.
- Initial facts: `Fever(ravi)`, `Cough(ravi)` and `BodyPain(ravi)`.
- Rules: `fever + cough → flu`; `flu + body pain → viral infection`; `viral infection → medical attention`.
- Target query: `ViralInfection(ravi)`.

Figure 1 — Smart Healthcare Knowledge Flow

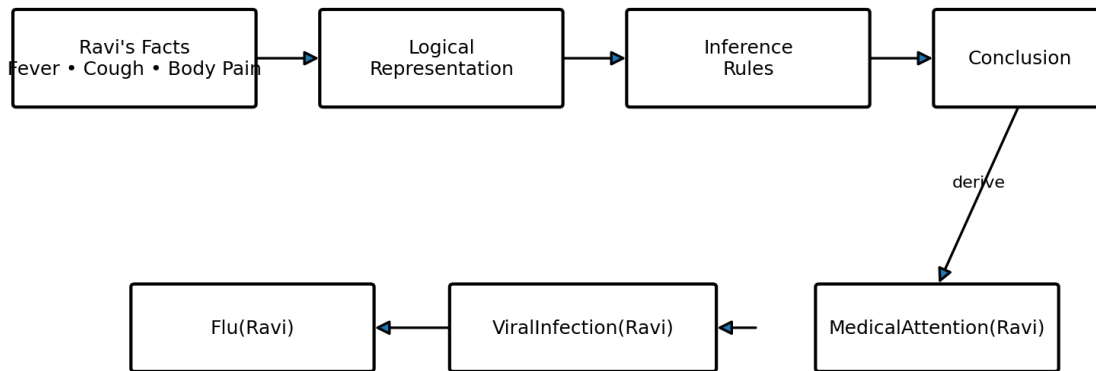


Figure 2. Problem analysis and transformation of the healthcare scenario into logical knowledge.

4. 2. Logical Representation and Mapping

Logical representation converts the natural-language statements into a formal structure that can be processed by an inference engine. For the fixed Ravi case, Propositional Logic can represent each statement with a symbol. For example, F can denote that Ravi has fever, C can denote cough, and B can denote body pain. The rules can then be written as implications such as $(F \wedge C) \rightarrow L$, where L represents flu.

First-Order Logic provides a more general representation because the rules are not restricted to Ravi. Predicates such as Fever(x), Cough(x), Flu(x) and ViralInfection(x) allow the same knowledge to be applied to any patient. Thus, the mapping moves from natural-language facts to formal predicates and from general medical reasoning statements to quantified rules. This representation makes the knowledge base structured, reusable and suitable for formal inference.

Representation Mapping

Natural-Language Knowledge	Formal Representation
Ravi has fever.	Fever(ravi)
Ravi has cough.	Cough(ravi)
Ravi has body pain.	BodyPain(ravi)
Patients with fever and cough may have flu.	$\forall x[(\text{Fever}(x) \wedge \text{Cough}(x)) \rightarrow \text{Flu}(x)]$
Patients with flu and body pain may have viral	$\forall x[(\text{Flu}(x) \wedge \text{BodyPain}(x)) \rightarrow \text{ViralInfection}(x)]$

infection.	
Patients with viral infection require medical attention.	$\forall x[\text{ViralInfection}(x) \rightarrow \text{MedicalAttention}(x)]$

Figure 2 – Knowledge Engineering Mapping

The following diagram shows the mapping of real-world knowledge (facts and rules) into logical predicates and first-order logic formulas used in the knowledge base.

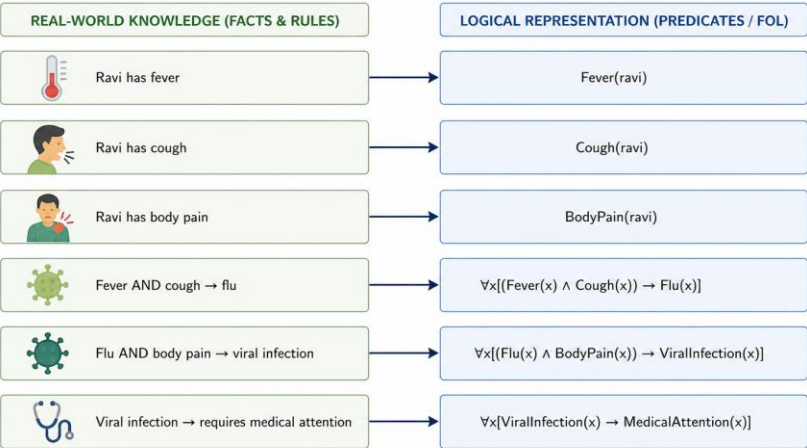


Figure 3. Knowledge engineering workflow showing natural-language statements mapped to formal logic.

5.3. Predicates, Variables, Constants and Quantifiers

First-Order Logic requires the identification of predicates, variables, constants and quantifiers so that the knowledge can describe individuals and their properties accurately. In this case, ravi is a constant because it refers to one particular patient. The variable x represents any patient in the domain. Predicates such as $\text{Fever}(x)$ and $\text{Cough}(x)$ describe properties of a patient, while $\text{Flu}(x)$ and $\text{ViralInfection}(x)$ represent possible conditions.

The universal quantifier $\forall$ is used because the supplied rules are intended to describe a general relationship between symptoms and conditions. For example, $\forall x[(\text{Fever}(x) \wedge \text{Cough}(x)) \rightarrow \text{Flu}(x)]$ means that for every patient x , if x has fever and cough, the rule allows flu to be inferred. This distinction between the constant ravi and the variable x is important for applying the general rules specifically to Ravi through substitution.

Component	Example	Meaning
Constant	ravi	Specific patient Ravi
Variable	x	Any patient in the domain
Predicate	Fever(x)	Patient x has fever
Predicate	ViralInfection(x)	Patient x has viral infection
Quantifier	$\forall x$	For every patient x

6. 4. Logical Connectives and Formula Construction

Logical connectives determine how individual statements are combined to form valid formulas. The conjunction operator $\wedge$ is used when two conditions must be satisfied together. The implication operator $\rightarrow$ connects a condition with its conclusion. Negation $\neg$ is used when a statement is assumed to be false, particularly during resolution by refutation. Correct use of these connectives is necessary to preserve the meaning of the original knowledge.

For example, the first rule is represented as $\text{Fever}(x) \wedge \text{Cough}(x) \rightarrow \text{Flu}(x)$. To prepare this implication for resolution, it is converted into the equivalent CNF clause $\neg \text{Fever}(x) \vee \neg \text{Cough}(x) \vee \text{Flu}(x)$. Similarly, the second rule becomes $\neg \text{Flu}(x) \vee \neg \text{BodyPain}(x) \vee \text{ViralInfection}(x)$. These formulas provide the logical structure required for formal proof.

Important Formula Constructions

$$\text{Fever}(x) \wedge \text{Cough}(x) \rightarrow \text{Flu}(x)$$

$$\neg \text{Fever}(x) \vee \neg \text{Cough}(x) \vee \text{Flu}(x)$$

$$\text{Flu}(x) \wedge \text{BodyPain}(x) \rightarrow \text{ViralInfection}(x)$$

$$\neg \text{Flu}(x) \vee \neg \text{BodyPain}(x) \vee \text{ViralInfection}(x)$$

$$\text{ViralInfection}(x) \rightarrow \text{MedicalAttention}(x)$$

$$\neg \text{ViralInfection}(x) \vee \text{MedicalAttention}(x)$$

7.5. Inference and Logical Reasoning

Inference is the process of deriving new knowledge from existing facts and rules. In this scenario, the system begins with three facts about Ravi: Fever(ravi), Cough(ravi) and BodyPain(ravi). The first rule can be applied because both fever and cough are available, producing Flu(ravi). The newly derived flu fact is then combined with BodyPain(ravi), allowing the second rule to derive ViralInfection(ravi).

The reasoning continues with the third rule. Once ViralInfection(ravi) is established, the system derives MedicalAttention(ravi). Therefore, the complete reasoning chain moves from symptoms to an intermediate condition, then to the final condition and the required action. The conclusion is not directly stored as an initial fact; it is produced through valid rule application.

Step	Known Information	New Inference
Initial	Fever(ravi), Cough(ravi), BodyPain(ravi)	Initial facts
1	Fever(ravi) $\wedge$ Cough(ravi)	Flu(ravi)
2	Flu(ravi) $\wedge$ BodyPain(ravi)	ViralInfection(ravi)
3	ViralInfection(ravi)	MedicalAttention(ravi)

Fever(ravi) + Cough(ravi) $\rightarrow$ Flu(ravi)

Flu(ravi) + BodyPain(ravi) $\rightarrow$ ViralInfection(ravi)

ViralInfection(ravi) $\rightarrow$ MedicalAttention(ravi)

Figure 4 — Complete Inference Sequence

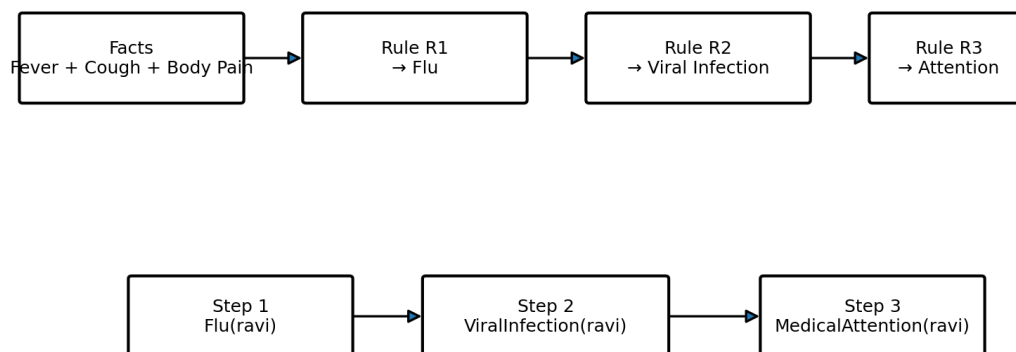


Figure 4.

Complete inference sequence from Ravi's symptoms to the final action.

8. 6. Unification and Substitution

Unification is the process of finding a substitution that makes two predicate expressions identical. It is necessary in this problem because the rules use the variable x , whereas the facts about the patient use the constant $ravi$. The substitution $\{x/ravi\}$ makes a general predicate such as $Fever(x)$ match the specific fact $Fever(ravi)$.

Two relevant examples can be demonstrated directly. First, $Fever(x)$ unifies with $Fever(ravi)$ using $\{x/ravi\}$. Second, $BodyPain(x)$ unifies with $BodyPain(ravi)$ using the same substitution. After substitution, the general rules become specific to Ravi, allowing the inference engine to apply them correctly. The predicate name and arity must match for the unification to succeed.

Unification Examples

Predicate Pair	Substitution	Result
$Fever(x)$ and $Fever(ravi)$	$\{x/ravi\}$	$Fever(ravi)$
$BodyPain(x)$ and $BodyPain(ravi)$	$\{x/ravi\}$	$BodyPain(ravi)$

$$\begin{aligned} & \forall x[(Fever(x) \wedge Cough(x)) \rightarrow Flu(x)] + \{x/ravi\} \\ & \Rightarrow (Fever(ravi) \wedge Cough(ravi)) \rightarrow Flu(ravi) \end{aligned}$$

9. 7. Resolution and Proof Construction

Resolution is a formal inference technique used here to prove the query $ViralInfection(ravi)$. The proof is constructed by contradiction: the query is negated and added to the knowledge base. The rules are first converted into CNF clauses. Resolution then combines clauses containing complementary literals until the empty clause is obtained.

The proof begins by resolving the fever and cough facts with the first rule to derive $Flu(ravi)$. This result is combined with the body-pain fact and the second rule to derive $ViralInfection(ravi)$. The derived statement contradicts the negated query $\neg ViralInfection(ravi)$. Resolving these two complementary statements produces the empty clause $\square$, showing that the negated query is inconsistent with the knowledge base.

Resolution Clauses

Clause	Expression
C1	Fever(ravi)
C2	Cough(ravi)
C3	BodyPain(ravi)
C4	$\neg\text{Fever}(x) \vee \neg\text{Cough}(x) \vee \text{Flu}(x)$
C5	$\neg\text{Flu}(x) \vee \neg\text{BodyPain}(x) \vee \text{ViralInfection}(x)$
C6	$\neg\text{ViralInfection}(\text{ravi})$

Proof Sequence

- Resolve C1 and C4 using $\{x/\text{ravi}\}$ to obtain $\neg\text{Cough}(\text{ravi}) \vee \text{Flu}(\text{ravi})$.
- Resolve the result with C2 to obtain $\text{Flu}(\text{ravi})$.
- Resolve $\text{Flu}(\text{ravi})$ with C5 using $\{x/\text{ravi}\}$ to obtain $\neg\text{BodyPain}(\text{ravi}) \vee \text{ViralInfection}(\text{ravi})$.
- Resolve the result with C3 to obtain $\text{ViralInfection}(\text{ravi})$.
- Resolve $\text{ViralInfection}(\text{ravi})$ with C6 to obtain the empty clause $\square$.

Figure 3 – Resolution Proof for ViralInfection(ravi)

The following diagram shows the step-by-step resolution proof to derive $\text{ViralInfection}(\text{ravi})$ using resolution rule until the empty clause ($\square$) is obtained.

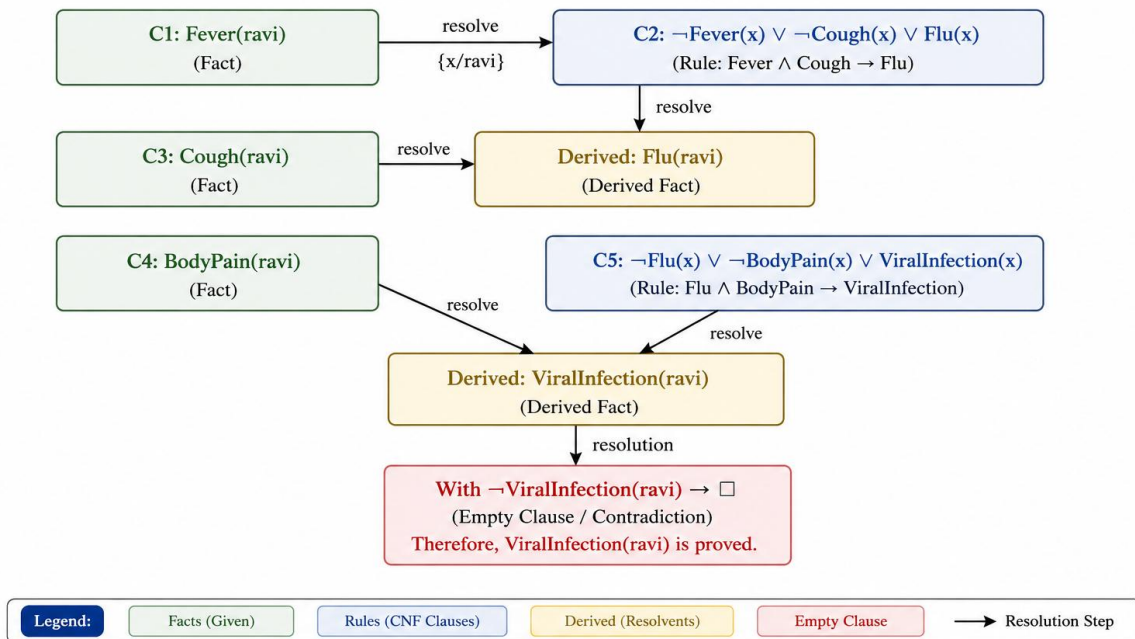


Figure 5. Resolution proof for Viral Infection(rave).

$$\text{KB} \cup \{\neg\text{Viral Infection}(\text{Ravi})\} =$$

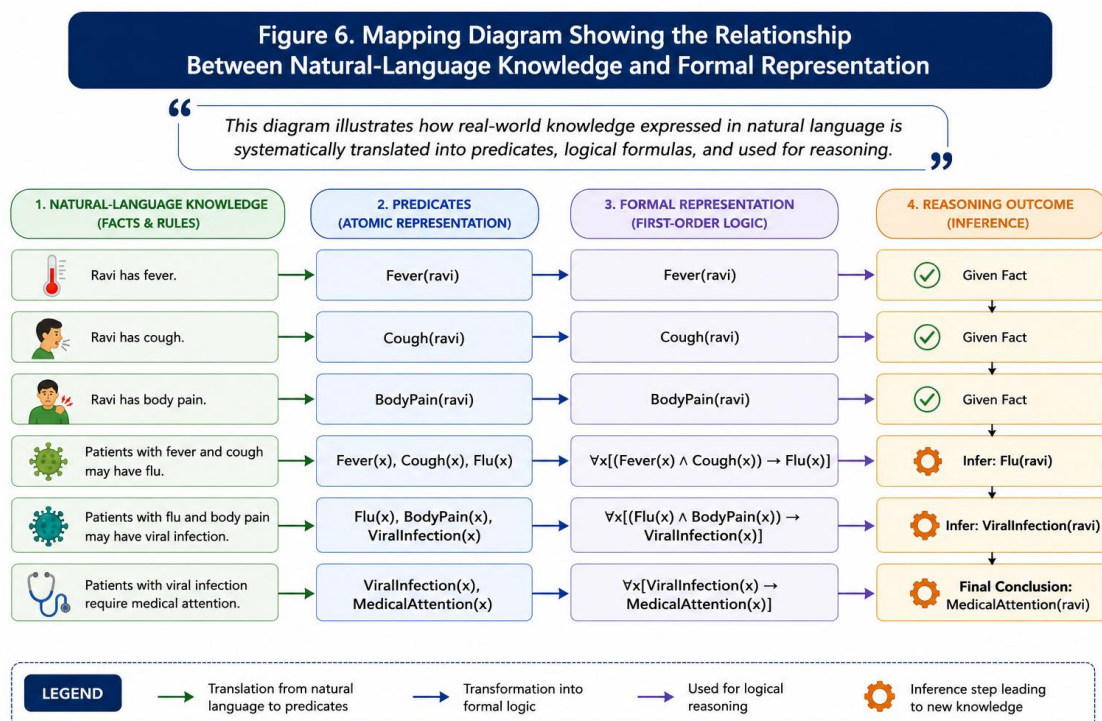
10. 8. Mapping Diagram / Clarity and Presentation

A mapping diagram provides a visual explanation of how the knowledge moves from the problem domain to formal reasoning. The main flow begins with Ravi's symptoms, which are represented as logical facts. The first rule produces flu, the second produces viral infection, and the third produces the required medical-attention conclusion. This visual structure helps the reader understand the dependency between each inference step.

Clarity is also maintained through consistent predicate names, a clear distinction between the constant ravi and variable x, tables for facts and rules, and separate sections for unification and resolution. The supporting diagrams show both the knowledge-engineering mapping and the proof process. These presentation choices make the formal reasoning easier to verify and directly support the assessment criterion.

Presentation Structure

- Natural language → logical facts and rules.
- Logical facts → inference through rule matching.
- General rules → specific conclusions through unification.
- CNF clauses → formal proof through resolution.
- Final conclusion → ViralInfection(ravi) and MedicalAttention(ravi).



11. COMPLETE CRITERION-WISE SUMMARY

Criterion	Weightage	Content Covered
1. Problem Analysis and Knowledge Identification	15%	Facts, entities, symptoms, rules and target query
2. Logical Representation and Mapping	20%	Propositional and FOL representations
3. Predicates, Variables, Constants and Quantifiers	15%	Predicates, variables, constants and universal quantifier
4. Logical Connectives and Formula Construction	10%	Conjunction, implication, negation and CNF
5. Inference and Logical Reasoning	15%	Step-by-step inference from symptoms to conclusion
6. Unification and Substitution	10%	Two relevant unification examples
7. Resolution and Proof Construction	10%	CNF clauses and refutation proof
8. Mapping Diagram / Clarity and Presentation	5%	Mapping diagrams, tables and structured presentation
Total	100%	

12. FINAL INFERENCE AND CONCLUSION

The knowledge representation system successfully converts Ravi's symptoms into formal logical facts and applies the supplied rules to derive new conclusions. Fever(ravi) and Cough(ravi) lead to Flu(ravi), while Flu(ravi) and BodyPain(ravi) lead to ViralInfection(ravi). The final rule then derives MedicalAttention(ravi).

The resolution proof confirms the target query by deriving the empty clause after assuming the negation of ViralInfection(ravi). Therefore, within the rules given in the academic scenario, Ravi is inferred to have a viral infection and consequently requires medical attention.

Note: This conclusion is the logical result of the educational knowledge base supplied in the question and is not intended to represent a real-world medical diagnosis.

EVALUATION RUBRICS:

Criterion	Weightage (%)	Excellent	Good	Average	Poor
Problem Understanding & Knowledge Identification	15%	13–15% – Clearly identifies entities, facts, relationships, assumptions, and reasoning requirements from the given scenario.	10–12% – Identifies most relevant knowledge with minor omissions.	7–9% – Identifies basic knowledge but misses several important elements.	0–6% – Poor or incorrect identification of the given knowledge.
Logical Representation & Mapping	20%	17–20% – Accurately maps real-world statements into appropriate propositional and first-order logic representations.	13–16% – Provides mostly correct logical mappings with minor errors.	9–12% – Provides partial mappings with several logical errors.	0–8% – Logical mapping is largely incorrect or missing.
Predicates, Variables, Constants & Quantifiers	15%	13–15% – Correctly identifies and applies predicates, variables, constants, universal and existential quantifiers.	10–12% – Uses the concepts correctly with minor errors.	7–9% – Demonstrates partial understanding with some incorrect usage.	0–6% – Concepts are incorrectly applied or missing.
Logical Connectives & Formula Construction	10%	9–10% – Correctly applies AND, OR, NOT, implication, and equivalence to	7–8% – Mostly correct use of connectives with minor errors.	5–6% – Basic use of connectives but several errors occur.	0–4% – Incorrect or incomplete formula construction.

		construct valid logical formulas.			
Inference & Logical Reasoning	15%	13–15% – Performs accurate step-by-step reasoning and derives valid conclusions from the given knowledge.	10–12% – Reasoning is mostly correct with minor gaps.	7–9% – Provides partial reasoning with some incorrect conclusions.	0–6% – Reasoning is incorrect, unsupported, or missing.
Unification & Substitution	10%	9–10% – Correctly identifies substitutions and Most General Unifier (MGU) and applies them accurately.	7–8% – Correctly performs most unification steps with minor errors.	5–6% – Demonstrates basic understanding but has several errors.	0–4% – Unable to perform unification correctly.
Resolution / Proof Construction	10%	9–10% – Correctly applies resolution and presents a clear, logically valid proof of the query.	7–8% – Resolution is mostly correct with minor errors.	5–6% – Shows partial understanding but proof contains gaps.	0–4% – Resolution is incorrect or not demonstrated.
Clarity, Mapping Diagram & Presentation	5%	5% – Provides a clear, well-organized mapping/logic diagram with accurate notation and excellent presentation.	4% – Presentation is clear with minor formatting or diagram issues.	3% – Basic presentation with limited clarity.	0–2% – Poorly organized, unclear, or incomplete presentation.